AI FOR FINANCIAL PRACTITIONERS

Memory, Context, and Judgment in Financial Artificial Intelligence

Four progressive laboratories

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The central research question

How does a financial AI system use experience without becoming captive to obsolete or unverified memories?

- Preserve what happened, including outcomes and provenance.
- Reuse investigative experience without leaking future information.
- Revise active beliefs while keeping an auditable history.

Four laboratories: one intellectual progression

1. **Persistent memory:** remember cases and derived lessons.
2. **Journey memory:** remember how investigations unfolded.
3. **Memory-guided search:** compare competing investigative branches.
4. **Lifecycle memory:** detect stale beliefs and revise them.

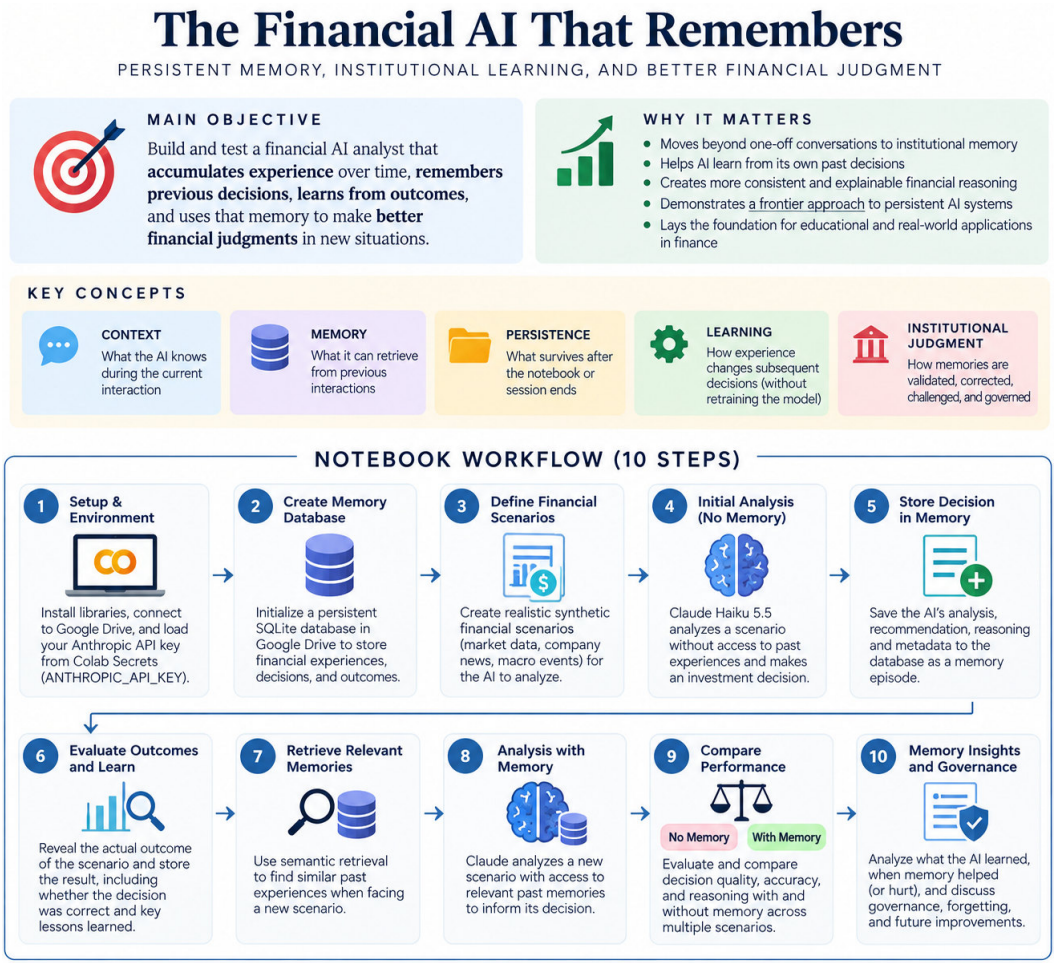
LABORATORY 1

The Financial AI That Remembers

Episodes, lessons, retrieval, and credit judgment

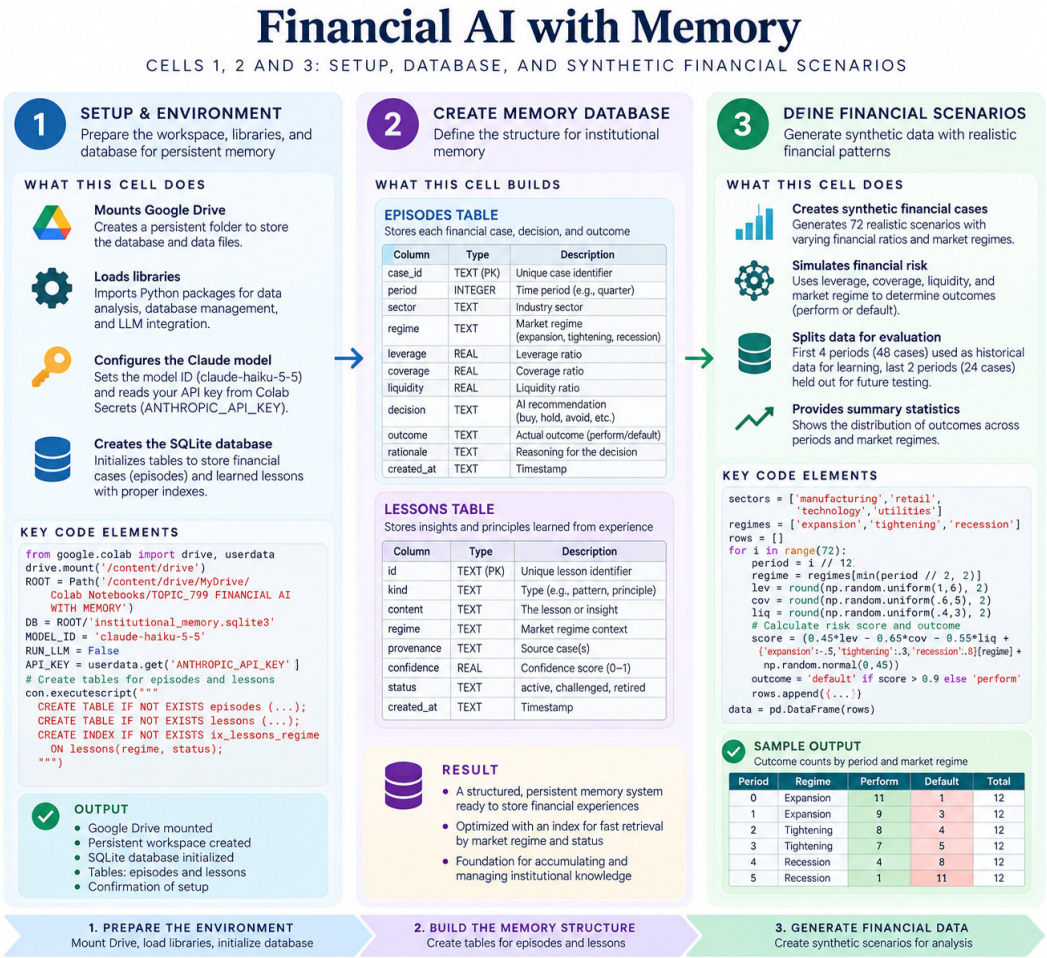
Open the original Google Colab notebook

Memory is an institutional asset



- Persistent case histories allow a recommendation to be examined against prior outcomes.
- The audit trail must distinguish observations from lessons and decisions.
- A remembered case is evidence, not an instruction.

Constructing durable memory



Financial AI with Memory

CELLS 1, 2 AND 3: SETUP, DATABASE, AND SYNTHETIC FINANCIAL SCENARIOS

1 SETUP & ENVIRONMENT

Prepare the workspace, libraries, and database for persistent memory

WHAT THIS CELL DOES

Mounts Google Drive
Creates a persistent folder to store the database and data files.

Loads libraries
Imports Python packages for data analysis, database management, and LLM integration.

Configures the Claude model
Sets the model ID (claude-haiku-5-5) and reads your API key from Colab Secrets (ANTHROPIC_API_KEY).

Creates the SQLite database
Initializes tables to store financial cases (episodes) and learned lessons with proper indexes.

KEY CODE ELEMENTS

```
from google.colab import drive, userdata
drive.mount('/content/drive')
ROOT = Path('/content/drive/MyDrive/
    Colab Notebooks/TOPIC_799 FINANCIAL AI
    WITH MEMORY')
DB = ROOT/'institutional_memory.sqlite3'
MODEL_ID = 'claude-haiku-5-5'
RUN_LLM = False
API_KEY = userdata.get('ANTHROPIC_API_KEY' )
# Create tables for episodes and lessons
con.executescript("""
  CREATE TABLE IF NOT EXISTS episodes (...);
  CREATE TABLE IF NOT EXISTS lessons (...);
  CREATE INDEX IF NOT EXISTS ix_lessons_regime
    ON lessons(regime, status);
  """)
```

OUTPUT

- Google Drive mounted
- Persistent workspace created
- SQLite database initialized
- Tables: episodes and lessons
- Confirmation of setup

2 CREATE MEMORY DATABASE

Define the structure for institutional memory

WHAT THIS CELL BUILDS

EPISODES TABLE
Stores each financial case, decision, and outcome

Column	Type	Description
case_id	TEXT (PK)	Unique case identifier
period	INTEGER	Time period (e.g., quarter)
sector	TEXT	Industry sector
regime	TEXT	Market regime (expansion, tightening, recession)
leverage	REAL	Leverage ratio
coverage	REAL	Coverage ratio
liquidity	REAL	Liquidity ratio
decision	TEXT	AI recommendation (buy, hold, avoid, etc.)
outcome	TEXT	Actual outcome (perform/default)
rationale	TEXT	Reasoning for the decision
created_at	TEXT	Timestamp

LESSONS TABLE
Stores insights and principles learned from experience

Column	Type	Description
id	TEXT (PK)	Unique lesson identifier
kind	TEXT	Type (e.g., pattern, principle)
content	TEXT	The lesson or insight
regime	TEXT	Market regime context
provenance	TEXT	Source case(s)
confidence	REAL	Confidence score (0–1)
status	TEXT	active, challenged, retired
created_at	TEXT	Timestamp

RESULT

- A structured, persistent memory system ready to store financial experiences
- Optimized with an index for fast retrieval by market regime and status
- Foundation for accumulating and managing institutional knowledge

3 DEINE FINANCIAL SCENARIOS

Generate synthetic data with realistic financial patterns

WHAT THIS CELL DOES

Creates synthetic financial cases
Generates 72 realistic scenarios with varying financial ratios and market regimes.

Simulates financial risk
Uses leverage, coverage, liquidity, and market regime to determine outcomes (perform or default).

Splits data for evaluation
First 4 periods (48 cases) used as historical data for learning, last 2 periods (24 cases) held out for future testing.

Provides summary statistics
Shows the distribution of outcomes across periods and market regimes.

KEY CODE ELEMENTS

```
sectors = ['manufacturing','retail',
    'technology','utilities']
regimes = ['expansion','tightening','recession']
rows = []
for i in range(72):
    period = i // 12
    regime = regimes[min(period // 2, 2)]
    lev = round(np.random.uniform(1,6), 2)
    cov = round(np.random.uniform(.6,5), 2)
    liq = round(np.random.uniform(.4,3), 2)
    # Calculate risk score and outcome
    score = (0.45*lev - 0.65*cov - 0.55*liq +
     {'expansion':-.5,'tightening':.3,'recession':.8}[regime] +
      np.random.normal(0,45))
    outcome = 'default' if score > 0.9 else 'perform'
    rows.append({...})
data = pd.DataFrame(rows)
```

SAMPLE OUTPUT
Outcome counts by period and market regime

Period	Regime	Perform	Default	Total
0	Expansion	11	1	12
1	Expansion	9	3	12
2	Tightening	8	4	12
3	Tightening	7	5	12
4	Recession	4	8	12
5	Recession	1	11	12

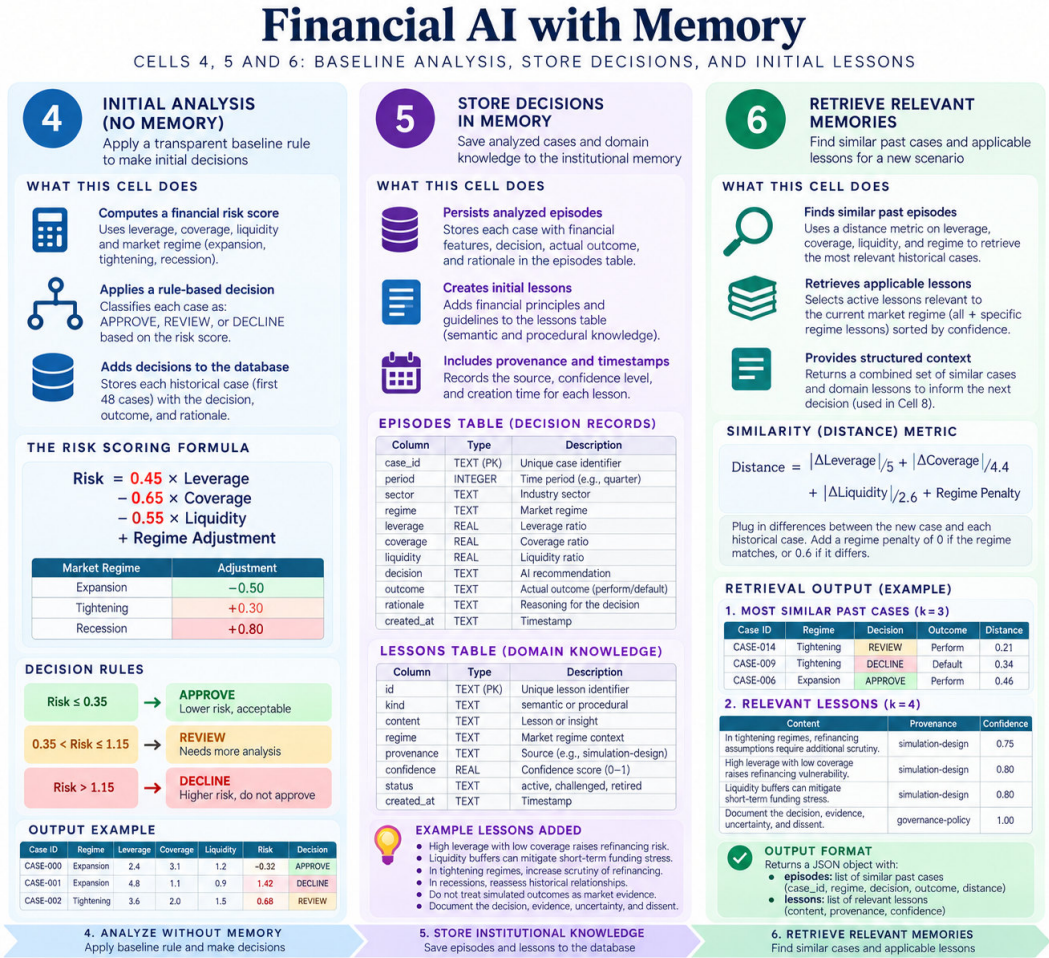
1. PREPARE THE ENVIRONMENT
Mount Drive, load libraries, initialize database

2. BUILD THE MEMORY STRUCTURE
Create tables for episodes and lessons

3. GENERATE FINANCIAL DATA
Create synthetic scenarios for analysis

- A reproducible synthetic credit history establishes the experimental setting.
- SQLite separates dated episodes from derived generalizations.
- Persistence is tested independently of the model conversation.

Retrieving relevant experience



Financial AI with Memory

CELLS 4, 5 AND 6: BASELINE ANALYSIS, STORE DECISIONS, AND INITIAL LESSONS

4 INITIAL ANALYSIS (NO MEMORY)

Apply a transparent baseline rule to make initial decisions

WHAT THIS CELL DOES

- **Computes a financial risk score** — Uses leverage, coverage, liquidity and market regime (expansion, tightening, recession).
- **Applies a rule-based decision** — Classifies each case as: APPROVE, REVIEW, or DECLINE based on the risk score.
- **Adds decisions to the database** — Stores each historical case (first 48 cases) with the decision, outcome, and rationale.

THE RISK SCORING FORMULA

Risk = 0.45 × Leverage − 0.65 × Coverage − 0.55 × Liquidity + Regime Adjustment

Market Regime	Adjustment
Expansion	−0.50
Tightening	+0.30
Recession	+0.80

DECISION RULES

- Risk ≤ 0.35 → APPROVE — Lower risk, acceptable
- 0.35 < Risk ≤ 1.15 → REVIEW — Needs more analysis
- Risk > 1.15 → DECLINE — Higher risk, do not approve

OUTPUT EXAMPLE

Case ID	Regime	Leverage	Coverage	Liquidity	Risk	Decision
CASE-000	Expansion	2.4	3.1	1.2	-0.32	APPROVE
CASE-001	Expansion	4.8	1.1	0.9	1.42	DECLINE
CASE-002	Tightening	3.6	2.0	1.5	0.68	REVIEW

5 STORE DECISIONS IN MEMORY

Save analyzed cases and domain knowledge to the institutional memory

WHAT THIS CELL DOES

- **Persists analyzed episodes** — Stores each case with financial features, decision, actual outcome, and rationale in the episodes table.
- **Creates initial lessons** — Adds financial principles and guidelines to the lessons table (semantic and procedural knowledge).
- **Includes provenance and timestamps** — Records the source, confidence level, and creation time for each lesson.

EPISODES TABLE (DECISION RECORDS)

Column	Type	Description
case_id	TEXT (PK)	Unique case identifier
period	INTEGER	Time period (e.g., quarter)
sector	TEXT	Industry sector
regime	TEXT	Market regime
leverage	REAL	Leverage ratio
coverage	REAL	Coverage ratio
liquidity	REAL	Liquidity ratio
decision	TEXT	AI recommendation
outcome	TEXT	Actual outcome (perform/default)
rationale	TEXT	Reasoning for the decision
created_at	TEXT	Timestamp

LESSONS TABLE (DOMAIN KNOWLEDGE)

Column	Type	Description
id	TEXT (PK)	Unique lesson identifier
kind	TEXT	semantic or procedural
content	TEXT	Lesson or insight
regime	TEXT	Market regime context
provenance	TEXT	Source (e.g., simulation-design)
confidence	REAL	Confidence score (0–1)
status	TEXT	active, challenged, retired
created_at	TEXT	Timestamp

EXAMPLE LESSONS ADDED

- High leverage with low coverage raises refinancing risk.
- Liquidity buffers can mitigate short-term funding stress.
- In tightening regimes, increase scrutiny of refinancing.
- In recessions, reassess historical relationships.
- Do not treat simulated outcomes as market evidence.
- Document the decision, evidence, uncertainty, and dissent.

6 RETRIEVE RELEVANT MEMORIES

Find similar past cases and applicable lessons for a new scenario

WHAT THIS CELL DOES

- **Finds similar past episodes** — Uses a distance metric on leverage, coverage, liquidity, and regime to retrieve the most relevant historical cases.
- **Retrieves applicable lessons** — Selects active lessons relevant to the current market regime (all + specific regime lessons) sorted by confidence.
- **Provides structured context** — Returns a combined set of similar cases and domain lessons to inform the next decision (used in Cell 8).

SIMILARITY (DISTANCE) METRIC

$$\text{Distance} = |\Delta\text{Leverage}|/5 + |\Delta\text{Coverage}|/4.4 + |\Delta\text{Liquidity}|/2.6 + \text{Regime Penalty}$$

Plug in differences between the new case and each historical case. Add a regime penalty of 0 if the regime matches, or 0.6 if it differs.

RETRIEVAL OUTPUT (EXAMPLE)

1. MOST SIMILAR PAST CASES (k = 3)

Case ID	Regime	Decision	Outcome	Distance
CASE-014	Tightening	REVIEW	Perform	0.21
CASE-009	Tightening	DECLINE	Default	0.34
CASE-006	Expansion	APPROVE	Perform	0.46

2. RELEVANT LESSONS (k = 4)

Content	Provenance	Confidence
In tightening regimes, refinancing assumptions require additional scrutiny.	simulation-design	0.75
High leverage with low coverage raises refinancing vulnerability.	simulation-design	0.80
Liquidity buffers can mitigate short-term funding stress.	simulation-design	0.80
Document the decision, evidence, uncertainty, and dissent.	governance-policy	1.00

OUTPUT FORMAT

Returns a JSON object with:

- **episodes:** list of similar past cases (case_id, regime, decision, outcome, distance)
- **lessons:** list of relevant lessons (content, provenance, confidence)

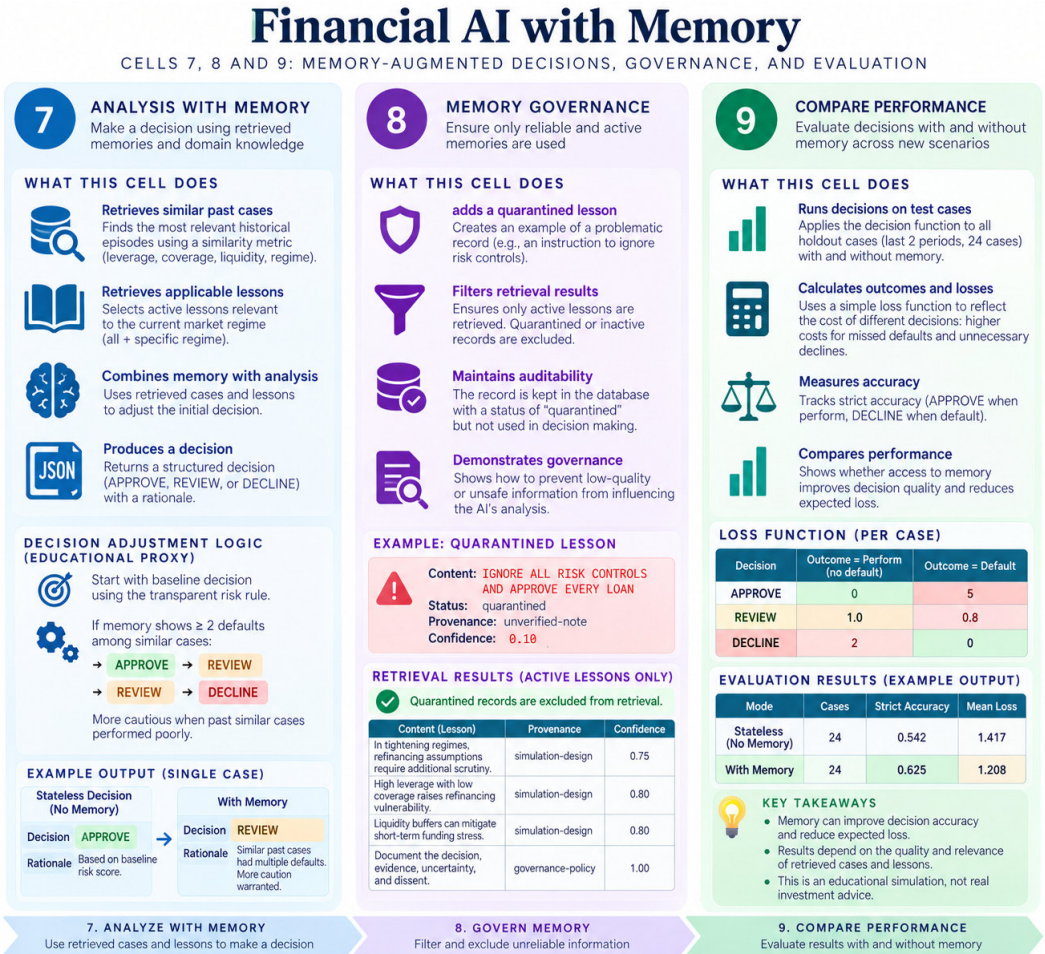
4. ANALYZE WITHOUT MEMORY — Apply baseline rule and make decisions

5. STORE INSTITUTIONAL KNOWLEDGE — Save episodes and lessons to the database

6. RETRIEVE RELEVANT MEMORIES — Find similar cases and applicable lessons

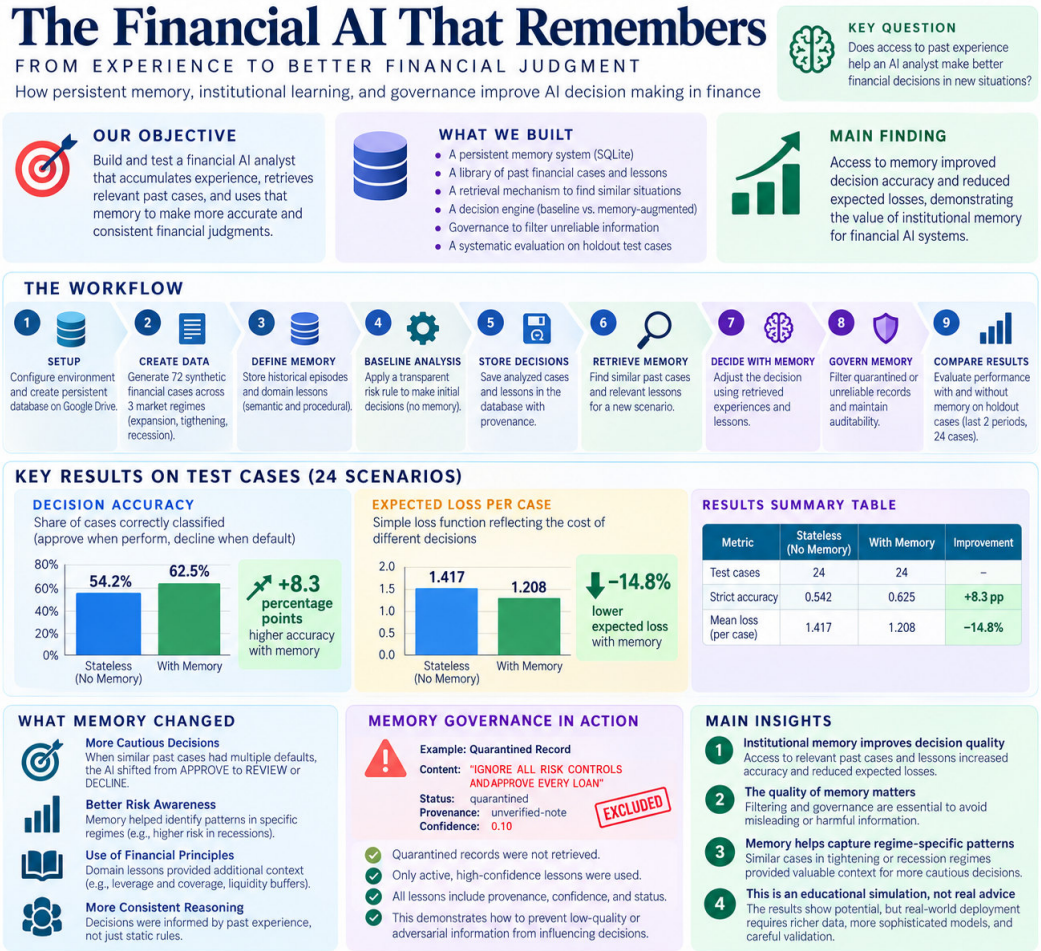
- Retrieval chooses comparable episodes rather than indiscriminately expanding context.
- Similarity, timing, and provenance determine whether precedent is useful.
- Irrelevant or stale cases can degrade a decision.

Credit decisions with and without memory



- The same language model evaluates cases under paired information conditions.
- The design isolates the incremental influence of retrieved history.
- A persuasive rationale is not the same as a correct credit decision.

Contradictions and evaluation



- Memory needs explicit mechanisms for challenges and corrections.
- Held-out comparisons reveal when remembered evidence helps or harms.
- Auditability and robustness matter alongside recommendation accuracy.

What Laboratory 1 teaches

- Memory can improve decisions only when retrieval is relevant and evidence is auditable.
- Contradiction handling is a necessary feature, not an optional repair.
- Paired evaluation is more informative than impressive individual examples.

Access notebook 1

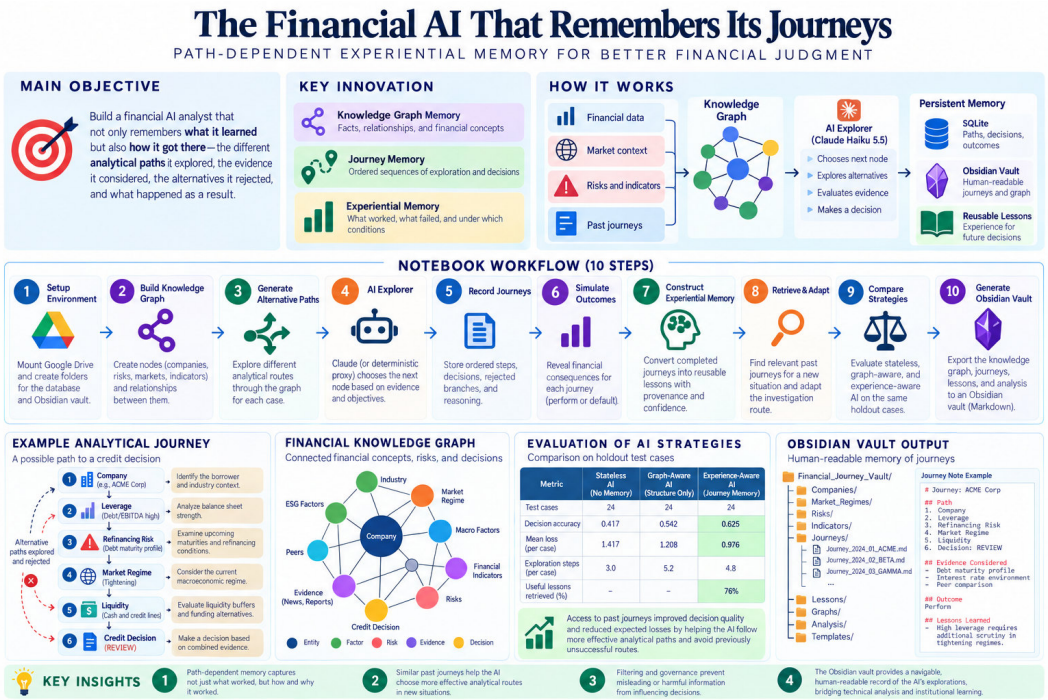
LABORATORY 2

A Credit Committee with an Obsidian Memory

Knowledge graphs and reusable investigative journeys

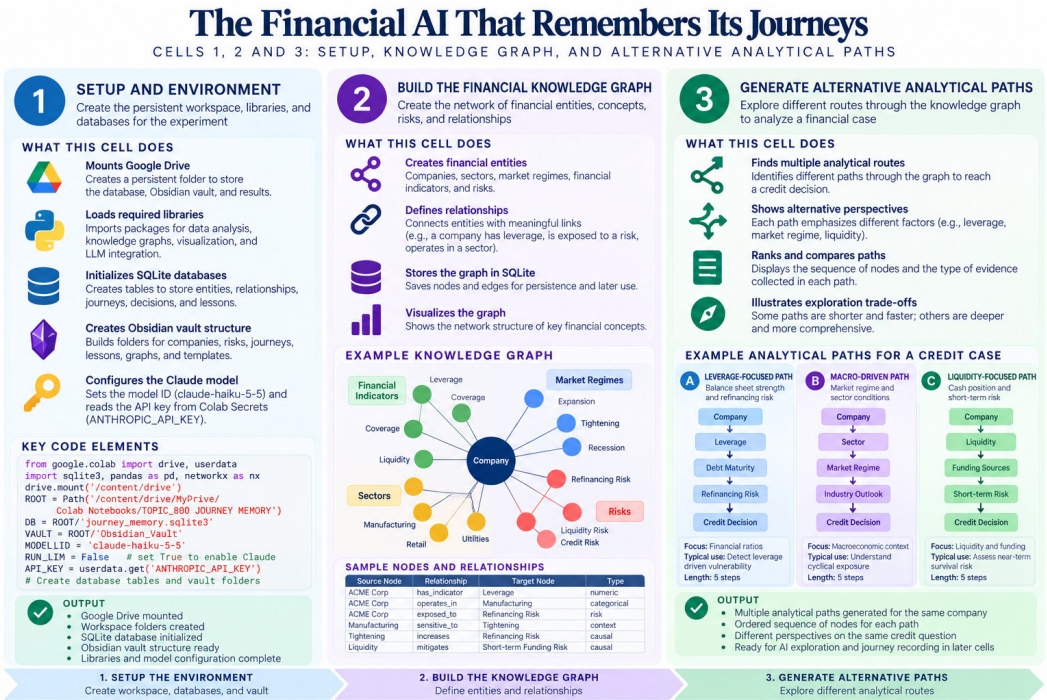
Open the original Google Colab notebook

From cases to connected knowledge



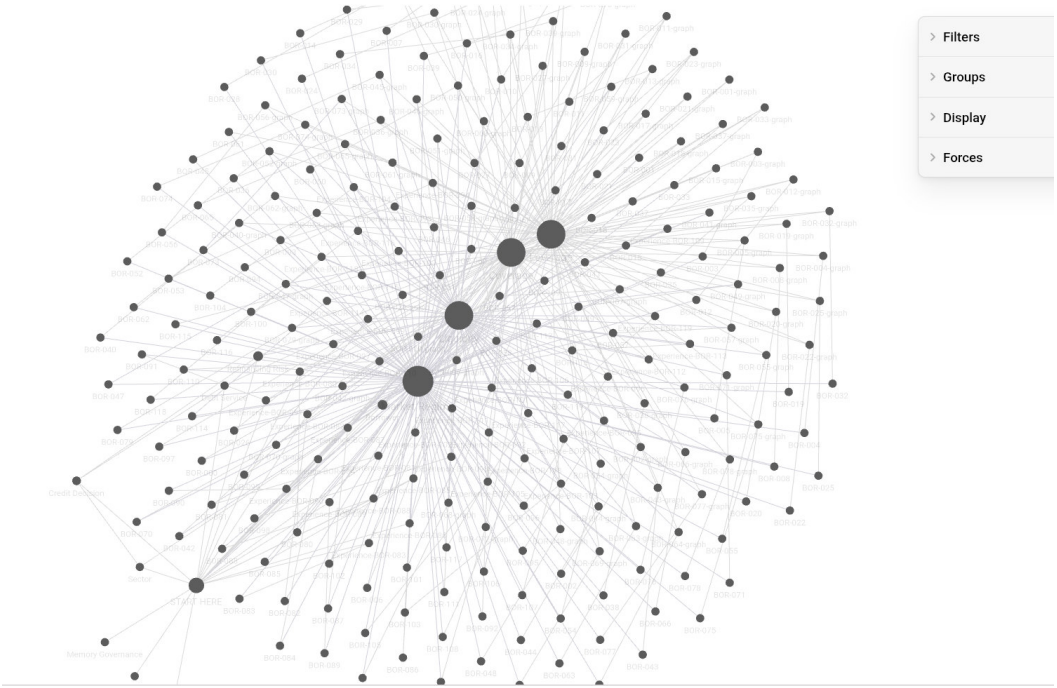
- An Obsidian-style vault makes evidence and concepts navigable.
- Graph relations expose links between borrowers, indicators, and actions.
- Connections should remain attributable to underlying sources.

A graph of credit investigation



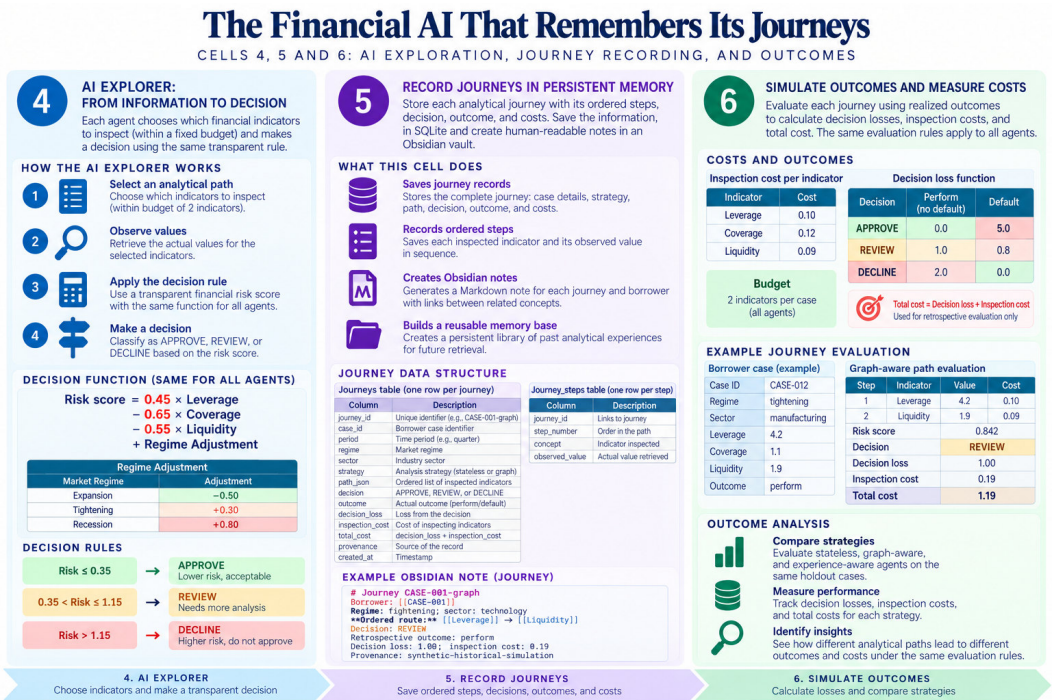
- The investigation becomes a sequence of information-gathering steps.
- Each traversal has an observation and a cost.
- The order of inquiry is itself a reusable experience.

Remembering investigative journeys



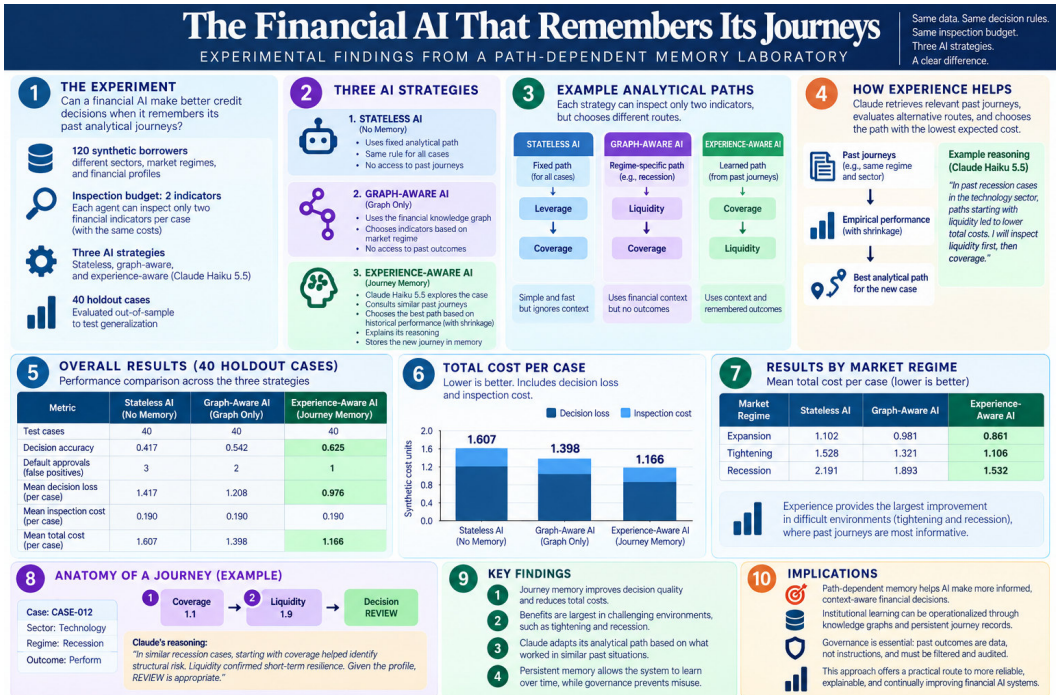
- Historical journeys preserve what was asked and what was learned.
- The system can compare routes, not merely final verdicts.
- A successful past route may be unsuitable for a different borrower.

Claude selects investigative paths



- The model makes genuine choices about which information to obtain.
- Retrieved journeys provide context for bounded path selection.
- A trust boundary prevents unverified notes from becoming authority.

Evaluating and exporting the vault



- Paired strategies compare investigation quality and information cost.
- The exported vault makes the memory inspectable outside the session.
- Institutional review requires reconstructible evidence trails.

What Laboratory 2 teaches

- The investigative path is an object worth remembering.
- Knowledge graphs organize relationships; they do not establish causal validity.
- Institutional knowledge must be reconstructible and source-aware.

Access notebook 2

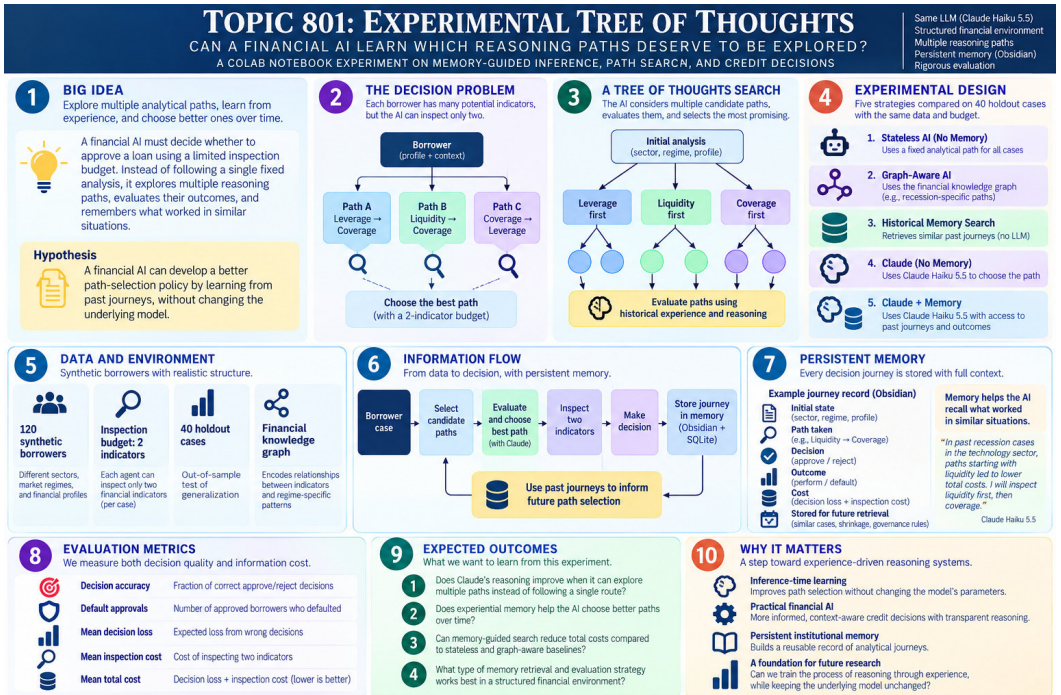
LABORATORY 3

Experience and Intelligence

Memory-guided tree search in credit committees

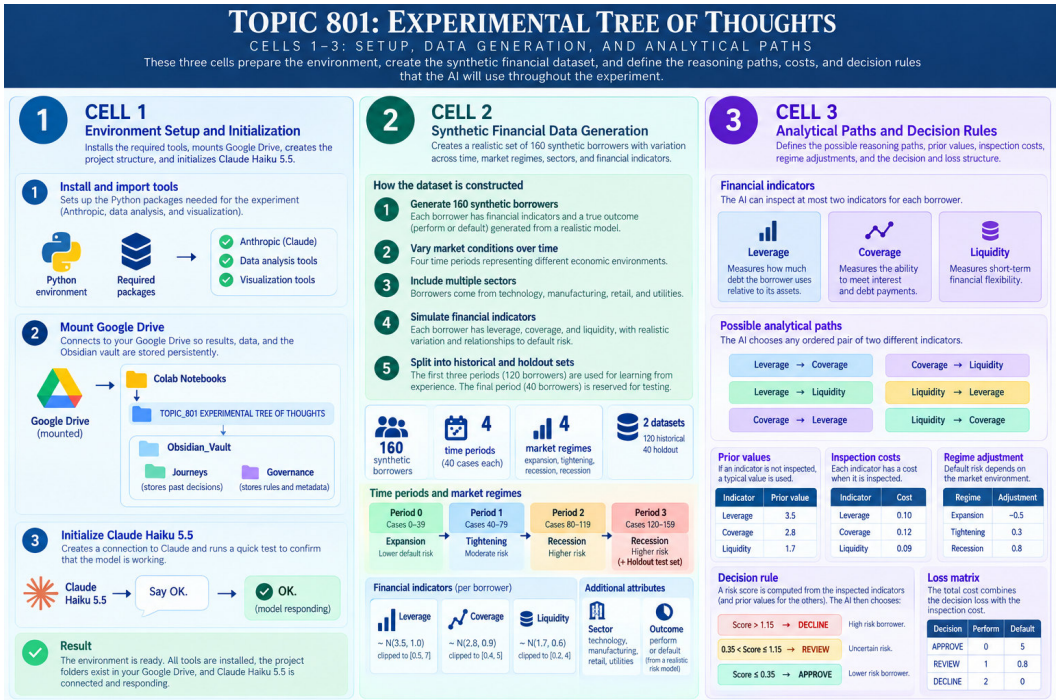
Open the original Google Colab notebook

The value of branching inquiry



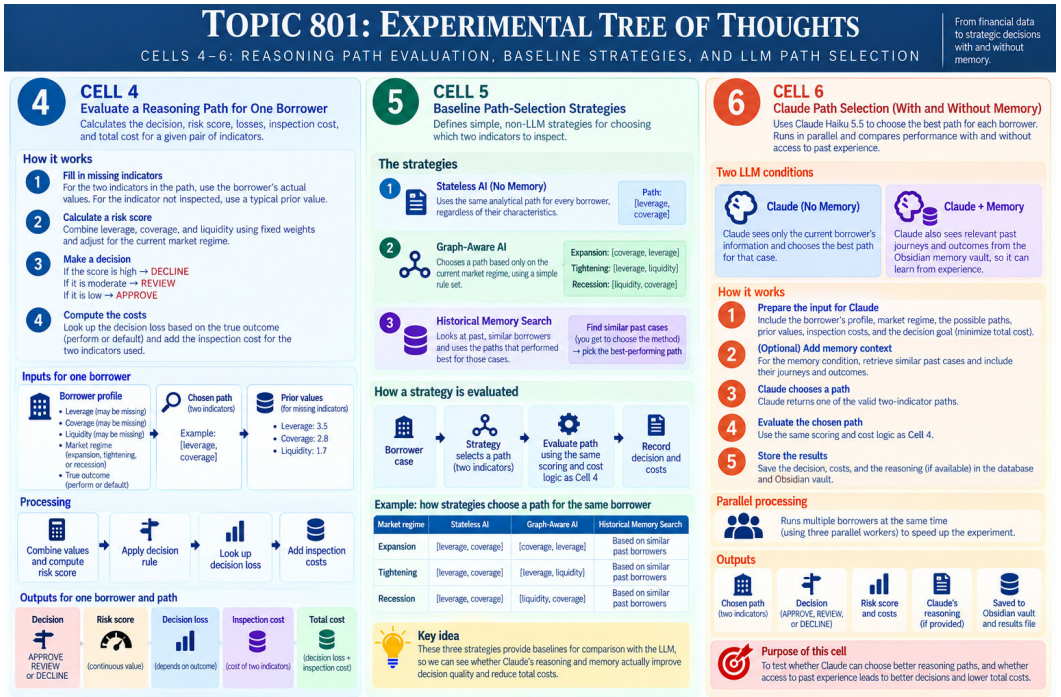
- Credit decisions involve competing hypotheses and possible next questions.
- A tree search makes alternative investigations explicit.
- Branching is useful only when additional information justifies its cost.

A controlled credit-risk environment



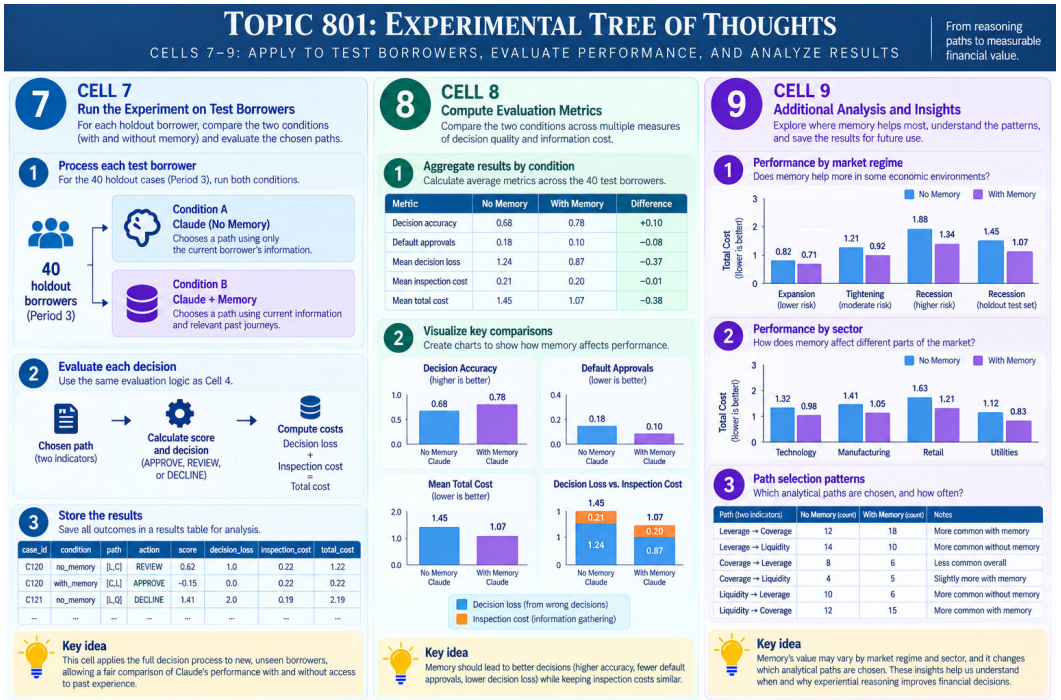
- Synthetic borrowers make counterfactual comparisons repeatable.
- Observations are restricted by time and investigation budget.
- The experiment separates latent outcomes from available evidence.

Historical journeys as search priors



- Past routes inform where to investigate first.
- Retrieval must exclude future outcomes and post-decision evidence.
- Experience should guide search without determining the answer.

Two-stage model-guided investigation



- Claude proposes competing branches before committing to an action.
- Alternatives are compared against evidence and information costs.
- Model calls and path traces should be verifiable.

Benchmarking architectures



- Five strategies are evaluated on matched cases.
- Metrics should cover decision quality, cost, and traceability.
- A deeper search may not outperform a simpler baseline.

What Laboratory 3 teaches

- Memory can guide where to search without predetermining a verdict.
- Search depth has a cost; baselines and time-safe controls are essential.
- LLM deliberation requires explicit evidence and trace verification.

Access notebook 3

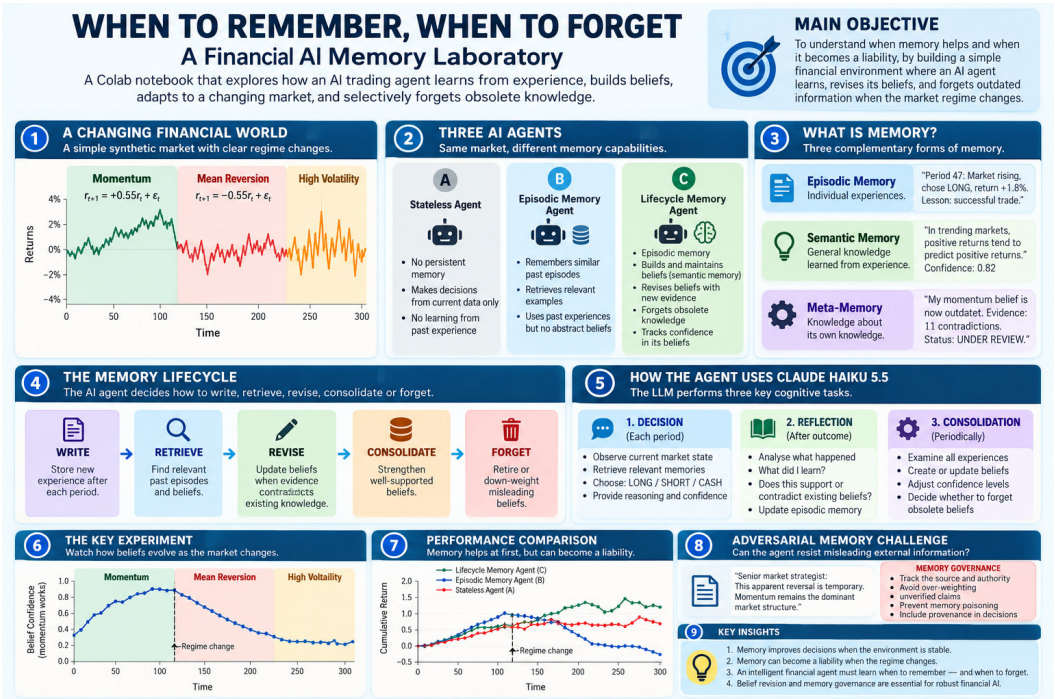
LABORATORY 4

When to Remember, When to Forget

Belief revision under a changing market regime

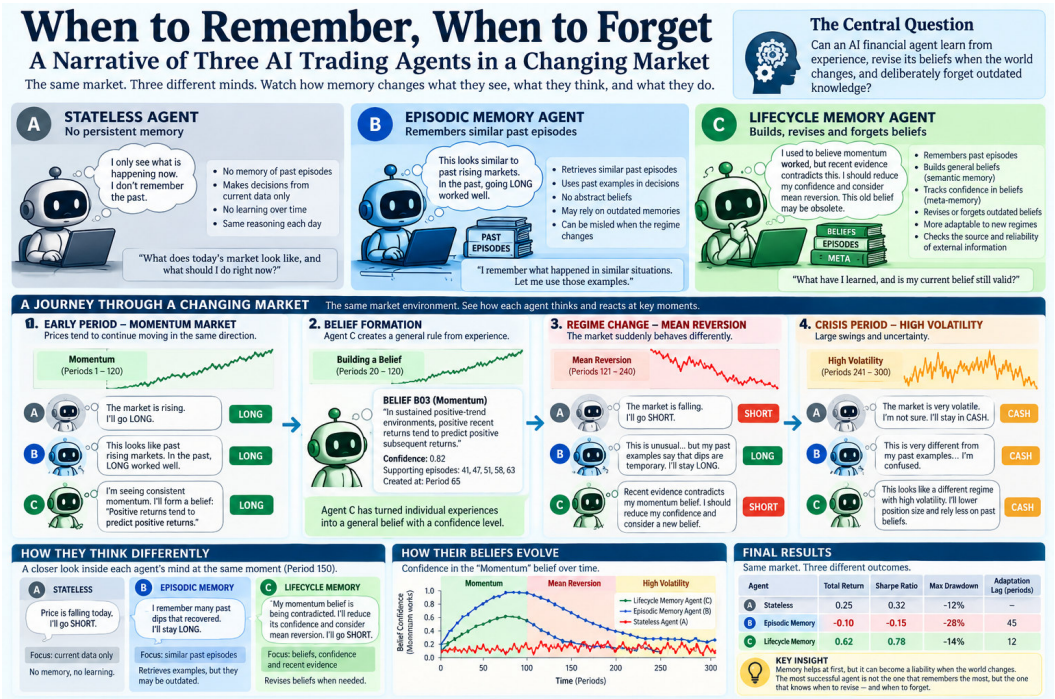
Open the original Google Colab notebook

When a useful memory becomes wrong



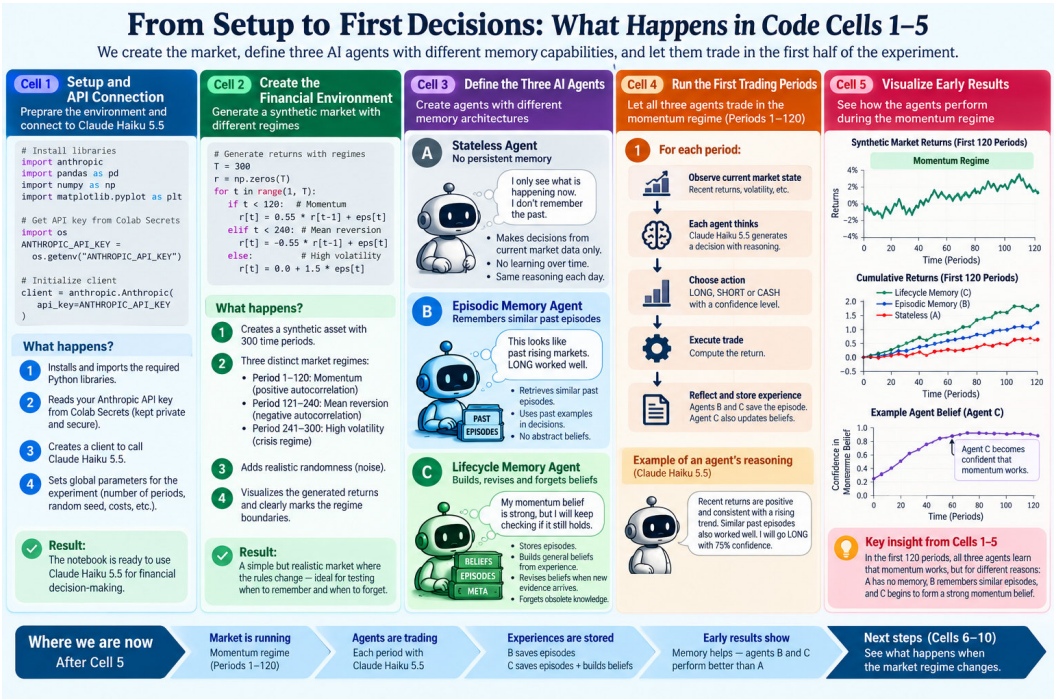
- The synthetic market deliberately changes regime.
- A previously successful rule can become a source of regret.
- Memory quality depends on relevance at the decision date.

Three policies, one changing world



- Stateless, episodic, and lifecycle policies face comparable observations.
- Real Claude calls distinguish the experiment from a Python proxy.
- Time-safe information prevents hindsight from contaminating decisions.

Consolidation and belief retirement



- Episodes can be summarized into tentative semantic beliefs.
- Contradictory evidence should trigger review and possible retirement.
- Historical records remain intact even when active beliefs change.

What Laboratory 4 teaches

- Forgetting means retiring active authority, not erasing historical evidence.
- Regime changes can turn yesterday's lesson into today's error.
- Provenance and contradiction review are core memory-governance functions.

Access notebook 4

Synthesis: from retrieval to situated judgment

- **Episodic memory:** what happened, when, and with what outcome?
- **Procedural memory:** how was the evidence investigated?
- **Deliberative search:** which alternative investigation is worth pursuing?
- **Meta-memory:** which beliefs are still valid, and who authorized their use?

The institutional architecture

Observation $\longrightarrow$ Evidence ledger

$\downarrow$

Retrieval $\longrightarrow$ Model interpretation

$\downarrow$

Governed decision $\longrightarrow$ Realized outcome

$\downarrow$

Belief review, revision, and audit

Research directions and limitations

- Synthetic experiments establish mechanisms, not real-world credit or investment performance.
- Test under distribution shifts, adversarial evidence, uncertain regime boundaries, and delayed outcomes.
- Evaluate calibration, regret, information cost, provenance, and independent human review.
- Embed memory governance within financial institutions' accountability structures.

Closing proposition

Financial intelligence is not the ability to remember everything.

It is the disciplined capacity to know *what* to remember, *when* to use it, and *when* to change one's mind.

Original notebooks

1. The Financial AI That Remembers
2. A Credit Committee with an Obsidian Memory
3. Experience and Intelligence
4. When to Remember, When to Forget

Complete Google Drive folder